Status	Finished
Started	Thursday, 9 October 2025, 12:00 PM
Completed	Thursday, 9 October 2025, 12:13 PM
Duration	13 mins 3 secs
Marks	2.00/3.00
Grade	**2.00** out of 3.00 (**66.68**%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade**.

The following rules apply:

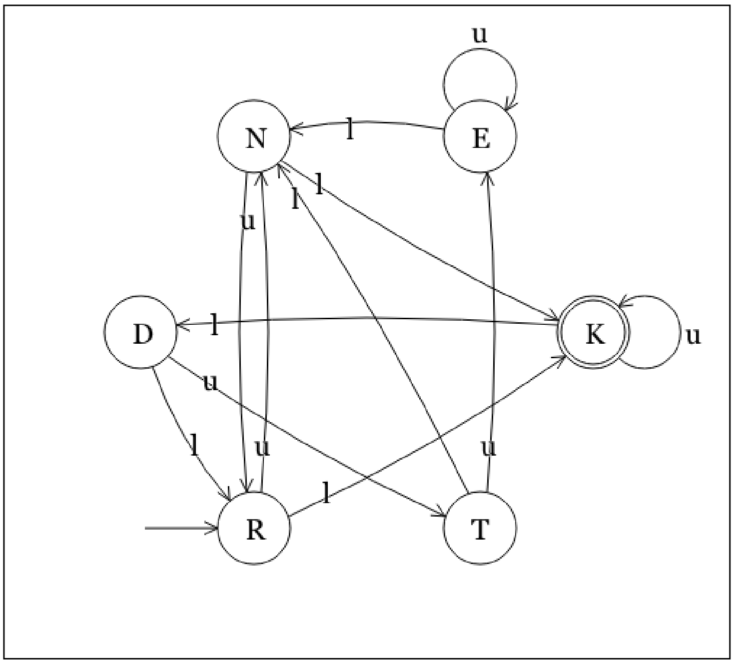
- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question **1**

Correct

Mark 1.00 out of 1.00

Minimize the following DFA.



Write the answer as a set of sets of equivalent states.

{{"K"},{"N","R"},{"E","D","T"}}

Your last answer was interpreted as follows:

$$\{\{K\},\{N,R\},\{E,D,T\}\}$$

Question **2**

Incorrect

Mark 0.00 out of 1.00

Consider the following specification.

{ m<13 } m,z := m',z' | z≥9

Which of the following implementations are correct with respect to the specification?

- [ ] { m=12 } m := 7; z := m+4 | z≥5
- [ ] { m=12 } m := 7; z := m+4 | z≥9
- [x] { m≤13 } m := 7; z := m+4 | z≥9
- [ ] { m<13 } m := 7; z := m+4 | z>9
- [x] { m≤13 } m := 7; z := m+4 | z≥5
- [x] { m<13 } m := 7; z := m+4 | z≥5
- [ ] { m≤13 } m := 7; z := m+4 | z>9
- [x] { m<13 } m := 7; z := m+4 | z≥9
- [ ] { m=12 } m := 7; z := m+4 | z>9
- [ ] None of the above

Question **3**

Correct

Mark 1.00 out of 1.00

Let R be a relation over $\{0,1,2,3,4,5,6,7,8,9\}$.

We know that R is an equivalence relation and a partial order.

Moreover, $(4,6)$ is not in R.

Select all pairs that are in R.

- [x] $[2,2]$
- [x] $[4,4]$
- [ ] $[4,5]$
- [ ] $[4,6]$
- [ ] $[5,2]$
- [ ] $[6,4]$
- [x] $[6,6]$
- [ ] $[7,1]$
- [ ] $[7,5]$
- [x] $[7,7]$
- [ ] $[8,7]$
- [ ] $[9,6]$
- [x] $[9,9]$
- [ ] None of the above

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 180

Signing code: 22648

Your last answer was interpreted as follows:

$$22648$$

◄ Test 1 (topics 1-3: Introduction, Concepts, Induction, Recursion, Grammars)

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